CL-1002
Programming Fundamentals
Lab # 16

Objectives:

- File Handling
- Understanding and Implementation of Constructs

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in C compiler/IDE and save source file **for each program.**
3. Do not copy from any source otherwise you will be penalized with negative marks.
4. Complete your lab **within given Time Slot**.
5. Add your source code in this word document + Make one ZIP file of your all source codes.
6. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO.
7. Submit your lab on Google Classroom.

Problem: 1 (Functions / String Manipulation)

Write a C++ program that implement a function named **SpecChar** which take char array/string an argument, and find the number of all special characters in the string.

Test Data:

send@gmail&&&com*#@22324^

Expected Output:

Count = 8

Special characters are:

@&&&*#@^

Problem: 2 (File Handling / Writing)

Write a C++ program that write in a file firstfile.txt your name, roll number and CGPA.

Example:
Name: XYZ
Roll Number: 21F-1111
CGPA:4

Problem: 3 (File Handling / Reading) (Marks = 2)

1. Write a C++ program that opens an already created file firstfile.txt (that you have created in above question).
2. Now read the data of file and display it. (Read the data until you reach end of file).
3. Now remove all the text from file.

Problem: 4 (File Handling / Reading & Writing) (Marks = 8)

You need to create your character dictionary we can called "Char Dict" using the concept of the file handling.

1. **Dictionary Initializations:**
   a. You need to generate 100 random characters both (Upper Case and Lower Case) and then you need to store in file, you can create text file and name as **dict.txt**
2. **Character Count**
   a. You need to create two function **myUpperCase** and **myLowerCase**.
   b. **myUpperCase** will read only Upper-Case Characters from the file and display.
   c. **myLowerCase** will read only Lower-Case Characters from the file and display.
3. **Update Dictionary**
   a. You need to ask from user if he/she want to update any character from the user.
   b. Let suppose user want to update 'C' character then in file all 'C' should be updated with the new character provided by the user.
4. **Display Content**
   a. You need to implement function that can show the whole data from the file, whenever user need.
5. **Dictionary Size**
   a. You need to create function that can display the size of whole dictionary to user.

Note:
In this task you need to make the UI/UX also in the best way, it is also included in grading.

Best Wishes